Lesson 19: Analysis of Paired Data

BSTA 551: Statistical Inference

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Lesson 19: Analysis of Paired Data

Review: Lesson 18

Two independent samples (Lesson 18):

- Two-sample z: σ_1^2, σ_2^2 known $\Rightarrow$ pivot $Z \sim N(0, 1)$
- **Welch t-test** (default): σ_1^2, σ_2^2 unknown $\Rightarrow T_W \approx t_{\hat{\nu}}$
- **Pooled t-test**: equal variances assumed $\Rightarrow T_P \sim t_{n_1+n_2-2}$ (exact)
- z-test for $p_1 - p_2$; F-test for σ_1^2/σ_2^2

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Today's Goals:

1. Recognize when data are **paired** vs. independent and why it matters
2. Show how pairing **reduces variance** when within-pair correlation is positive
3. Derive and apply the **paired t-test** and **paired CI** for μ_D (DBC 10.3)
4. Understand the link between paired analysis and one-sample inference
5. Apply to crossover trials, pre/post studies, and matched case-control designs

Roadmap

Design	Key Operation	Method	Reference
Pre/post (same subjects)	$D_i = X_{1i} - X_{2i}$	One-sample t on $\{D_i\}$	DBC 10.3
Crossover trial	Same	Paired t	DBC 10.3
Matched case-control	Same	Paired t	DBC 10.3
Bilateral measurements	Same	Paired t	DBC 10.3

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Motivating Example

Crossover trial: 20 hypertensive patients each receive an antihypertensive drug for 4 weeks, then (after washout) placebo for 4 weeks, in random order. Outcome: systolic blood pressure reduction (mmHg). Each patient is their own control so the two measurements are **not independent**; they are paired within each patient.

The Paired Design

When Are Data Paired?

Data are paired when each observation in group 1 is **naturally linked** to exactly one observation in group 2:

Design	Example
Pre/post (same subject)	SBP before vs. after 12 weeks of treatment
Crossover trial	Same patient receives drug and placebo in random order
Matched case-control	Cases matched 1:1 to controls on age and sex
Repeated measurement	Lab value at baseline and 6-month follow-up
Bilateral organs	Right vs. left eye intraocular pressure

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The Critical Distinction

Paired: n pairs $\rightarrow$ n differences $\rightarrow$ **one-sample problem on D_i**

Independent: $n_1 + n_2$ separate observations $\rightarrow$ **two-sample problem**

Applying a two-sample test to paired data **ignores within-pair correlation**, discards information, and loses power — sometimes substantially.

The Paired Reduction

Setup: $(X_{1i}, X_{2i}), i = 1, \ldots, n$ are n independent pairs (but within each pair, X_{1i} and X_{2i} are correlated).

Define the **within-pair difference**: $D_i = X_{1i} - X_{2i}$

Then $D_1, \ldots, D_n \overset{iid}{\sim}$ some distribution with:

$$\mu_D = E[D_i] = \mu_1 - \mu_2 \qquad \sigma_D^2 = \mathrm{Var}(D_i)$$

Everything reduces to one-sample inference on $D_1, \ldots, D_n$.

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Why the variance is smaller: Let $\rho = \mathrm{Cor}(X_{1i}, X_{2i})$. Then:

$$\sigma_D^2 = \mathrm{Var}(X_{1i} - X_{2i}) = \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2$$

When $\rho > 0$ (the typical case in biological paired designs):

$$\sigma_D^2 = \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2 \; < \; \sigma_1^2 + \sigma_2^2$$

Pairing **removes between-subject noise** by differencing within each pair.

Paired t-Test (DBC 10.3)

Let $\bar{D} = \frac{1}{n}\sum_{i=1}^{n} D_i$ and $S_D^2 = \frac{1}{n-1}\sum_{i=1}^{n}(D_i - \bar{D})^2$.

Paired t-statistic for $H_0 : \mu_D = \mu_0$ (usually $\mu_0 = 0$):

$$T = \frac{\bar{D} - \mu_0}{S_D/\sqrt{n}} \sim t_{n-1}$$

Exact under normality of D_i; approximately valid by CLT for large n.

Paired $100(1-\alpha)\%$ CI for μ_D:

$$\bar{D} \;\pm\; t_{\alpha/2,\, n-1} \cdot \frac{S_D}{\sqrt{n}}$$

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In R:

```
t.test(x1, x2, paired = TRUE)   # from two measurement vectors
t.test(d)                        # from pre-computed differences
```

Note

The paired t-test is **identical** to the one-sample t-test on the differences — all the theory from Lessons 7–8 applies directly.

Variance Reduction: How Much Does Pairing Help?

When $\sigma_1 = \sigma_2 = \sigma$, the SE of $\bar{D}$ in the paired analysis is:

$$\mathrm{SE}_{\text{paired}} = \frac{\sigma_D}{\sqrt{n}} = \frac{\sigma\sqrt{2(1-\rho)}}{\sqrt{n}}$$

The SE from an unpaired analysis with the same total n subjects per group would be:

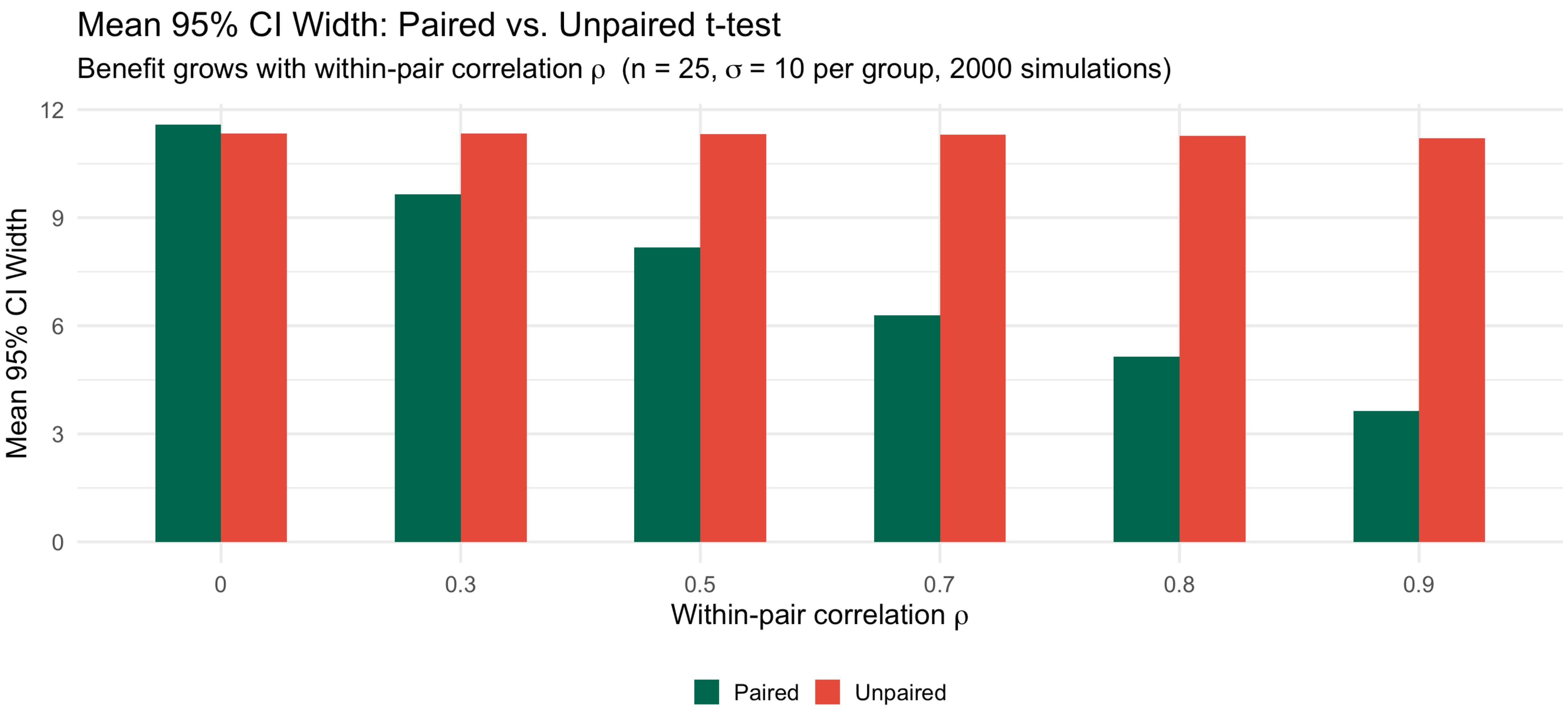
$$\mathrm{SE}_{\text{unpaired}} = \sigma\sqrt{\frac{2}{n}}$$

The ratio of CI widths (paired to unpaired) is $\sqrt{1-\rho}$.

```r
tibble(rho = c(0, 0.3, 0.5, 0.7, 0.8, 0.9)) |>
  mutate(
    width_ratio     = round(sqrt(1 - rho), 3),
    pct_narrower    = scales::percent(1 - sqrt(1 - rho), accuracy = 1)
  ) |>
  knitr::kable(
    col.names = c("rho", "CI width ratio (paired / unpaired)",
                  "% CI width reduction"),
    caption   = "Benefit of pairing when sigma_1 = sigma_2"
```

Visualizing Variance Reduction by Simulation

▶ Code



Medical Example: Crossover Trial

Crossover Trial: Antihypertensive Drug vs. Placebo

Study (DBC 10.3 style): 20 hypertensive patients complete a crossover trial. Each receives both an antihypertensive drug and placebo (random order, 4-week washout). Outcome: systolic BP reduction (mmHg). Test $H_0 : \mu_D = 0$ vs. $H_a : \mu_D < 0$ (drug lowers BP more than placebo) at $\alpha = 0.05$.

```r
set.seed(551)
n   <- 20
rho <- 0.75
L   <- chol(10^2 * matrix(c(1, rho, rho, 1), 2))
# Drug arm lowers BP ~5 mmHg more than placebo
xy  <- matrix(rnorm(2 * n), n, 2) %*% L +
       matrix(c(0, -5), n, 2, byrow = TRUE)

sbp_placebo <- xy[, 1]
sbp_drug    <- xy[, 2]
d           <- sbp_drug - sbp_placebo  # negative = drug better

cat(sprintf("Mean D (drug - placebo):  %.2f mmHg\n", mean(d)))
```

```
Mean D (drug - placebo):  -6.22 mmHg
```

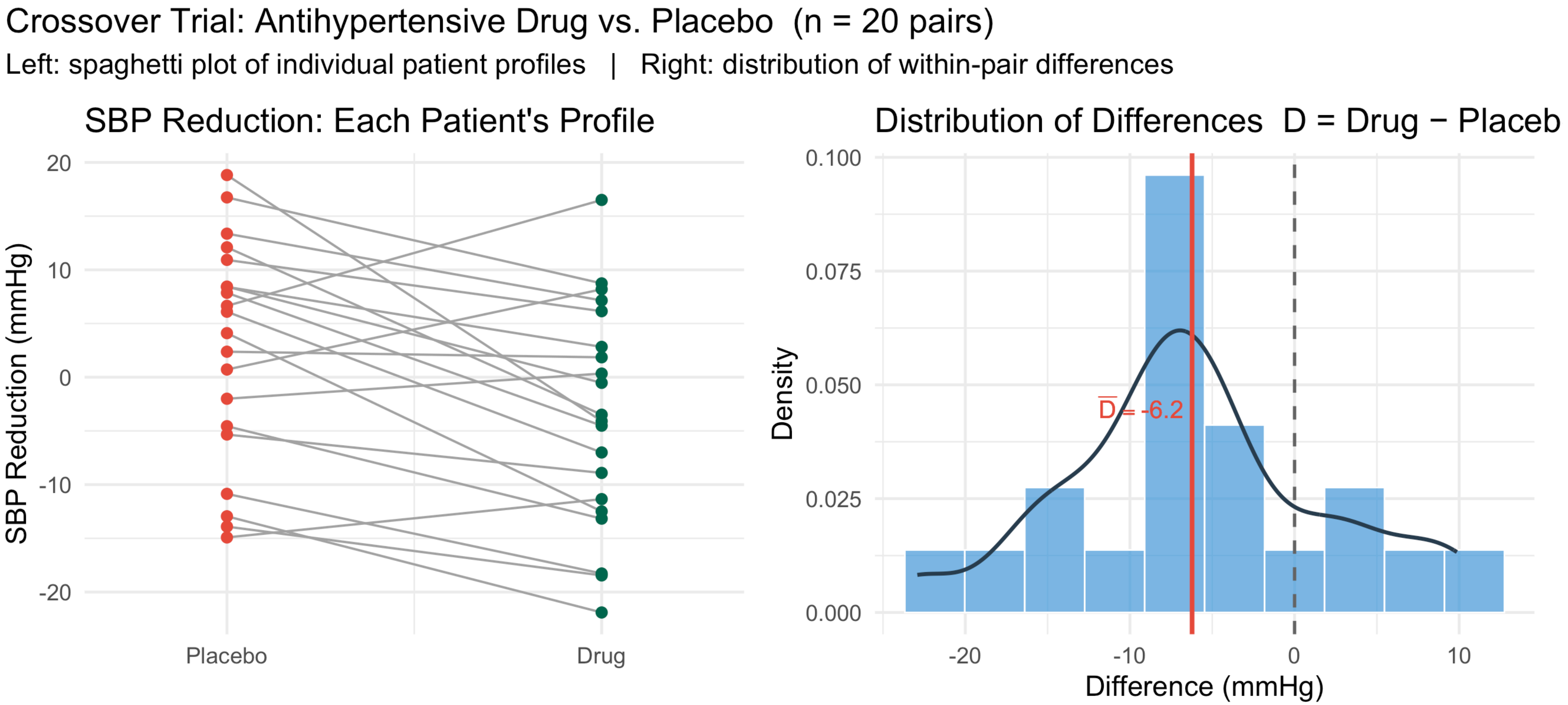
```r
cat(sprintf("SD(D):                    %.2f mmHg\n", sd(d)))
```

```
SD(D):                    8.09 mmHg
```

```r
result <- t.test(sbp_drug, sbp_placebo, paired = TRUE, alternative = "less")
tidy(result)
```

Visualizing the Paired Data

▶ Code



What If We Incorrectly Used a Two-Sample Test?

```r
paired_res   <- t.test(sbp_drug, sbp_placebo, paired = TRUE, alternative = "less")
unpaired_res <- t.test(sbp_drug, sbp_placebo, paired = FALSE, alternative = "less")

cat("--- Paired t-test (CORRECT) ---\n")
```

```
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```

```r
cat(sprintf("  t = %.3f,  df = %d,  p = %.5f\n",
            paired_res$statistic,
            paired_res$parameter,
            paired_res$p.value))
```

```
  t = -3.440,  df = 19,  p = 0.00137
```

```r
cat(sprintf("  95%% CI: (%.2f, %.2f)\n\n",
            paired_res$conf.int[1], paired_res$conf.int[2]))
```

```
  95% CI: (-Inf, -3.10)
```

```r
cat("--- Welch two-sample t-test (INCORRECT for paired data) ---\n")
```

```
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```

```r
cat(sprintf("  t = %.3f,  df = %.1f,  p = %.5f\n",
            unpaired_res$statistic,
            unpaired_res$parameter,
            unpaired_res$p.value))
```

```
  t = -1.907,  df = 38.0,  p = 0.03207
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Pre/Post Study Example

Pre/Post: Ferritin After Iron Supplementation

Study (DBC 10.3): 12 iron-deficient patients have serum ferritin (ng/mL) measured before and after 8 weeks of oral iron supplementation. Test for a significant increase ($H_a : \mu_D > 0$).

```r
ferritin_pre  <- c( 8, 11,  9, 14,  7, 12,  6, 15, 10,  9, 13,  8)
ferritin_post <- ferritin_pre + rnorm(length(ferritin_pre), 4, 3)
d_iron        <- ferritin_post - ferritin_pre

cat(sprintf("Mean pre:   %.1f   |   Mean post:  %.1f\n",
            mean(ferritin_pre), mean(ferritin_post)))
```

```
Mean pre:   10.2   |   Mean post:  15.8
```

```r
cat(sprintf("Mean D:     %.1f   |   SD(D):      %.1f   |   n = %d\n",
            mean(d_iron), sd(d_iron), length(d_iron)))
```

```
Mean D:     5.6   |   SD(D):      2.3   |   n = 12
```

```r
res_iron <- t.test(ferritin_post, ferritin_pre,
                   paired = TRUE, alternative = "greater")
tidy(res_iron)
```

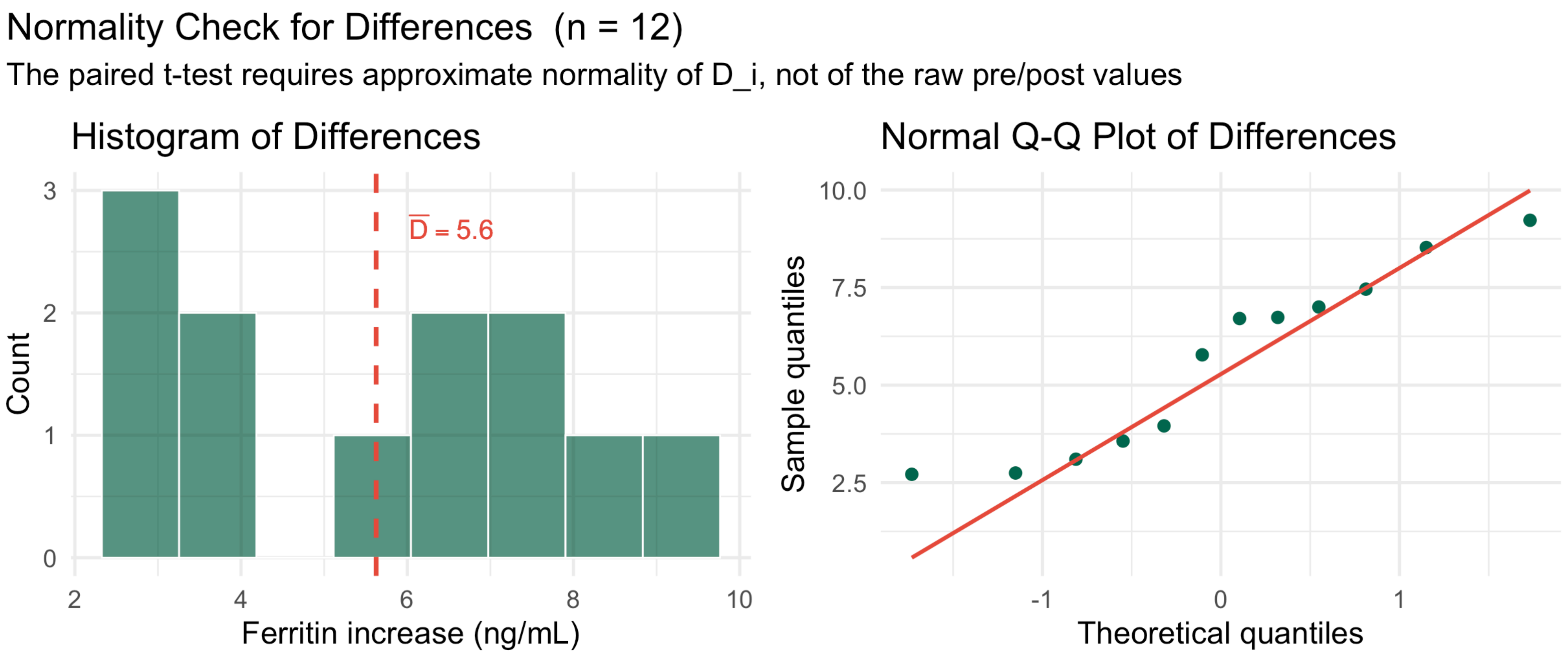
```
# A tibble: 1 × 8
  estimate statistic    p.value parameter conf.low conf.high method  alternative
     <dbl>     <dbl>      <dbl>     <dbl>    <dbl>     <dbl> <chr>   <chr>
1     5.63      8.40 0.00000204        11     4.42       Inf Paired… greater
```

```r
cat(sprintf("One-sided 95%% CI: (%.2f, Inf)\n", res_iron$conf.int[1]))
```

```
One-sided 95% CI: (4.42, Inf)
```

Checking the Normality Assumption for D_i

▶ Code



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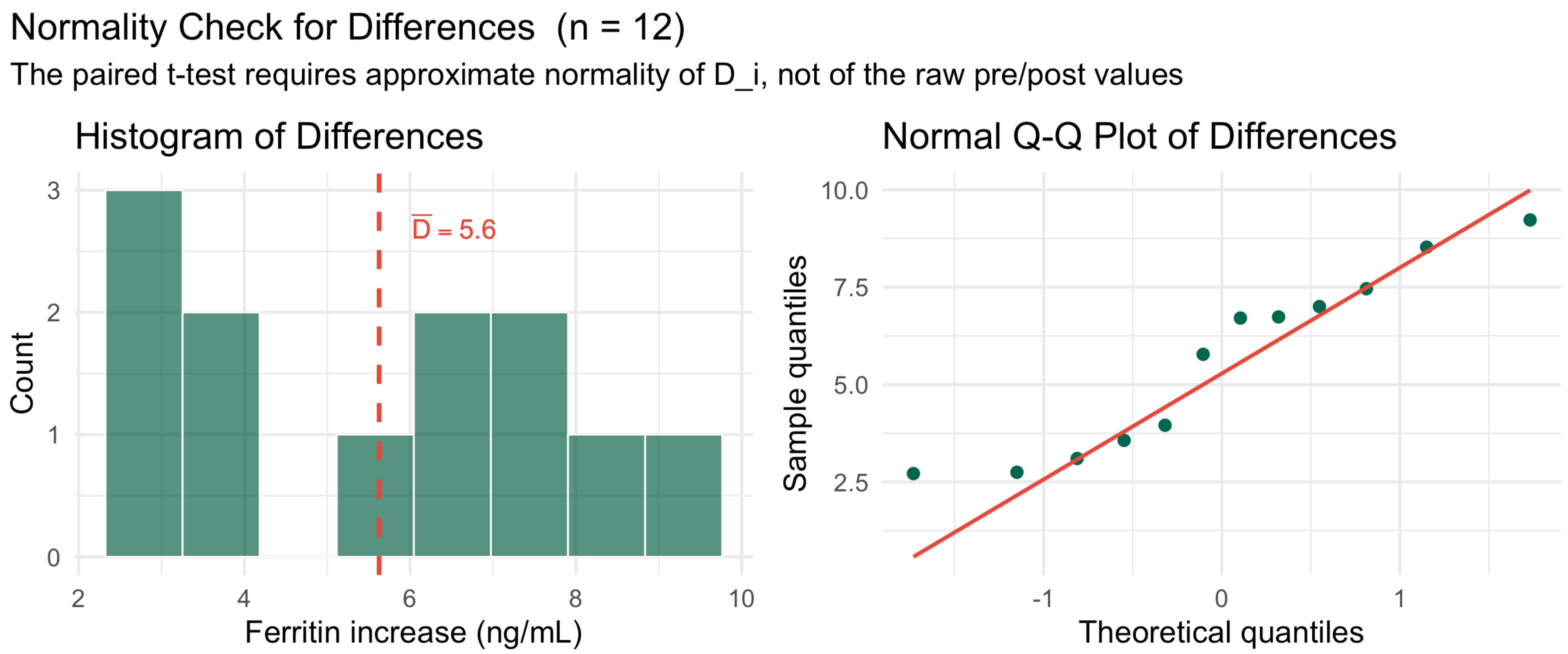

With $n = 12$, normality of D_i matters. The Q-Q plot looks reasonable here. For small n with visibly non-normal differences, use the **Wilcoxon signed-rank test** or **sign test** (Lesson 21).

Matched Case-Control Design

Matched Pairs: Smoking and Lung Cancer

Study: 15 lung cancer cases are each individually matched to a healthy control on age (± 2 yr) and sex. Exposure: pack-years of smoking. Within each matched pair, compute the case–control difference.

```r
set.seed(7)
cases    <- round(pmax(0, rnorm(15, mean = 35, sd = 15)))
controls <- round(pmax(0, cases - rnorm(15, mean = 20, sd = 12)))
d_match  <- cases - controls

cat("Pack-years - Cases:    ", cases,    "\n")
```

```
Pack-years - Cases:    69 17 25 29 20 21 46 33 37 68 40 76 69 40 63
```

```r
cat("Pack-years - Controls:", controls, "\n")
```

```
Pack-years - Controls: 43 8 9 9 0 0 18 0 34 33 18 47 42 32 46
```

```r
cat(sprintf("\nMean difference (case - control): %.1f  |  SD: %.1f\n",
            mean(d_match), sd(d_match)))
```

```

Mean difference (case - control): 20.9  |  SD: 9.2
```

```r
t.test(cases, controls, paired = TRUE) |> tidy()
```

```
# A tibble: 1 × 8
  estimate statistic     p.value parameter conf.low conf.high method alternative
     <dbl>     <dbl>       <dbl>     <dbl>    <dbl>     <dbl> <chr>  <chr>
1     20.9      8.78 0.000000459        14     15.8      26.0 Paire… two.sided
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Paired Analysis Checklist

Before choosing paired vs. two-sample analysis, confirm:

1. **1-to-1 link exists**: same subject measured twice, or two subjects deliberately matched
2. **Within-pair correlation is positive**: almost always true for biological measurements on the same individual
3. n **= number of pairs**, not total observations — paired t-test has $n-1$ df, not $2n-2$
4. D_i **approximately normal** (or n large enough for CLT)

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⚠ **Common Mistake**

Reporting "$n_1 = n_2 = 20$" for a pre/post study and running `t.test(pre, post)` without `paired = TRUE`. This treats $2n = 40$ observations as independent, underestimates the SE, and produces an invalid test. The correct n is 20 pairs, df = 19.

Key Takeaways

Key Takeaways (DBC 10.3)

Paired Data: Main Points

1. **Paired designs** arise from pre/post measurements, crossover trials, bilateral organs, and matched studies; each pair yields one difference $D_i = X_{1i} - X_{2i}$
2. **Variance reduction:** $\mathrm{Var}(D_i) = \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2$; with $\sigma_1 = \sigma_2 = \sigma$, the CI width ratio (paired/unpaired) is $\sqrt{1-\rho}$ — pairing eliminates between-subject noise when $\rho > 0$
3. **Paired t-test:** $T = \bar{D}/(S_D/\sqrt{n}) \sim t_{n-1}$; identical to one-sample t on the D_i; exact under normality of D_i, approximately valid by CLT for moderate n
4. **Always use `paired = TRUE`** in R when the design is paired — omitting it ignores correlation, widens CIs, and reduces power
5. **Check normality of D_i** (not of X_{1i} or X_{2i}) with a histogram and Q-Q plot; for small n with non-normal D_i, use nonparametric alternatives (Lesson 21)

Looking Ahead

Next class (Lesson 20): Bootstrap and Permutation Methods for Two Samples (C&H Ch. 3, 5; DBC 10.6)

- Bootstrap CI for $\mu_1 - \mu_2$: resample each group independently
- Permutation test: shuffle labels under $H_0 : F_1 = F_2$
- Paired bootstrap: resample differences
- When do resampling methods outperform parametric tests?

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- Permutation test: shuffle labels under $H_0 : F_1 = F_2$
- Paired bootstrap: resample differences
- When do resampling methods outperform parametric tests?

Lesson 21: Nonparametric inference — sign test, Wilcoxon signed-rank, Wilcoxon rank-sum, asymptotic relative efficiency (DBC Ch. 14)

Lesson 22: Course Review

References

- Devore, Berk, and Carlton. *Modern Mathematical Statistics with Applications* (Springer). Section 10.3.
- Chihara, L. M. and Hesterberg, T. C. *Mathematical Statistics with Resampling and R* (Wiley). Chapter 3.

Questions?

Thank you!